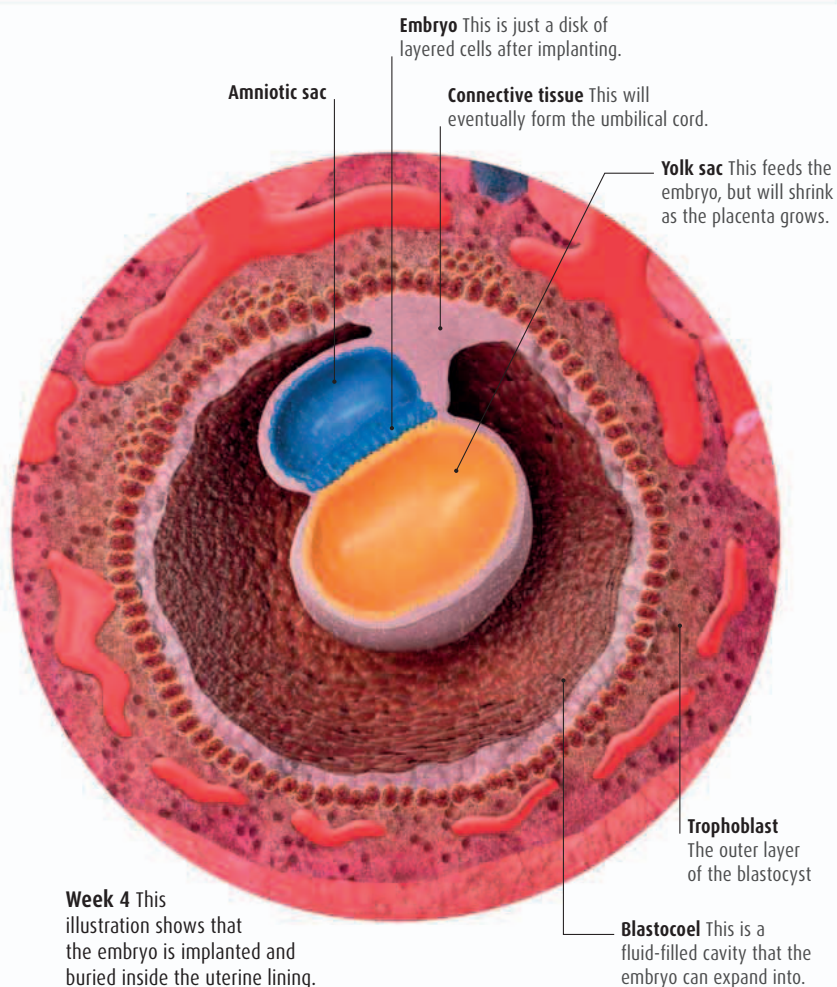


## Q How are the fertilized cells nourished before there's a placenta?

The early embryo is fed by a “yolk sac,” which develops from some of the cells of the blastocyst.

Despite being called a yolk sac, there's hardly any yolk in it, but what is there provides enough energy and nutrition to see your developing baby through until the fully formed placenta takes over at around 12 weeks. The placenta doesn't begin to form until you're about four weeks pregnant (which is about two weeks after conception, since pregnancy is dated from the first day of your last menstrual period).

The early embryo is just a disk of cells; on one side there is an amniotic sac, and on the other side is the yolk sac. A basic circulatory system draws nutrients from the yolk sac and delivers them into the developing embryo cells. As the embryo grows it is enveloped by the amniotic sac, while the yolk sac remains outside, but connected to the embryo. Part of the yolk sac then triggers cells in the blastocyst to begin making the placenta. The yolk sac slowly shrinks as the placenta becomes stronger, disappearing into the umbilical cord when the placenta is fully formed. It is while the embryo is dependent on the yolk sac that it is most vulnerable—this is one of the reasons why the risk of miscarriage is highest in the first trimester and becomes lower as the placenta begins to share the job of sustaining the baby.



## Q Does it matter where the cells implant in my uterus?

**No and yes.** The most important thing is that the bundle of cells finds a comfortable home in your uterine lining, rather than anywhere else, such as the fallopian tubes. (If this happens, your pregnancy is termed ectopic (see p.308), which can be life threatening for you.) Usually, your smart body draws the blastocyst toward the top (the superior wall) and back (the posterior wall) of your uterus. This is the best place for implantation because when the placenta grows, it's very unlikely to block your baby's eventual exit through your cervix. Sometimes, though, the blastocyst implants lower down in your uterus and the placenta forms too close to the cervix (this is called placenta previa; see p.147). This can happen if, for example, you have uterine fibroids (see p.146) or a scar from a previous surgery (such as a cesarean section) that make it difficult for

the blastocyst to implant. Although your baby can develop healthily wherever in the uterus the blastocyst makes its home, the place where it implants can have consequences for your labor and your experience of birth.

## Q How does my baby develop from just layers of cells?

**At five weeks, the embryo is made up of three layers of cells that develop into different parts of your baby's body.** These layers undergo a complex process of three-dimensional folding to form the basic bodily structures. The ectoderm is the outermost layer of the embryo and will form, among other things, your baby's nervous system, his facial features (including his eyes, nose, mouth, and ears), and his skin, nails, and hair. It also becomes your baby's anal canal. The mesoderm is the middle layer, and, will become, among other things, your baby's circulatory

system, skeletal system (including bones, cartilage, muscles, and ligaments), blood cells and bone marrow, kidneys, and reproductive organs. The endoderm is the innermost layer of the embryo and will become your baby's respiratory system, digestive system, and bladder, as well as several important glands of his endocrine (hormone) system, including the thyroid, parathyroid, and thymus glands.

**100–150**  
cells are present in the **blastocyst** (as a result of cell division) by the time it enters the uterine cavity, ready to implant.